No Edge to the Halo: KiDS-1000 Galaxy Lensing Against a Dark-Energy-Tied Gravity Framework's Cold Halos

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Abstract

In a gravity framework where the galactic acceleration scale is set by the dark-energy density, $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$ (the $\frac{1}{2}$ is fitted), each galaxy's extra lensing mass is a collisionless "cold" component. That component settles onto the law $g_N = g^2/(g+a_0)$. We test four pre-registered statements about it against the KiDS-1000 lensing rotation curves of isolated galaxies (Brouwer et al. 2021; 4 stellar-mass bins, full covariance).

1. **The edge is excluded.** A halo that stops where it has used up the galaxy's own cosmic share of cold energy ($5.364\,M_b$, at $5.84\,r_M$) is excluded: $\Delta\chi^2 = +197$ within 0.31 Mpc and $+753$ over all radii, on both a_0 footings.
2. **The framework's own infall repairs it.** The framework's collisionless infall fills onto the law out to 20–$50\,r_M$, and fits as well as an edgeless law ($\Delta\chi^2 = +0.1$), provided each galaxy draws on $N \geq 3$–4 times its own baryons.
3. **The supply needs no fit inside 0.31 Mpc.** Taking $N = 1/f_{\rm ret}$ from the baryon census used in the framework's simulations ($N \approx 6$–10 at these masses) fits within 0.31 Mpc with no fitted parameter ($\Delta\chi^2 = -0.2$ and -1.5 for the two footings).
4. **Neighbours narrow the outer gap.** Beyond each catchment edge (0.32–0.45 Mpc), adding a two-halo term from the framework's own billion-particle simulation narrows the outer deficit from $\Delta\chi^2 = +149$ to $+11$, with an effective bias of 0.2–0.5.

The absolute fit is poor for every model, including the plain law ($\chi^2 \approx 160$ on 52 degrees of freedom). The misfit sits in the most massive bins: the known massive and early-type lensing excess.

1 Setup

- **Settled profile.** For a galaxy of baryonic mass M_b, the settled cold mass is

$$M_c(r) = \tfrac{1}{2}M_b\left(\sqrt{1+4r^2/r_M^2}-1\right), \qquad r_M = \sqrt{GM_b/a_0}.$$

  Far out it behaves as an isothermal sphere with locked amplitude, $\Delta\Sigma \to \sqrt{GM_b a_0}/(4GR)$.
- **Data.** The KiDS-1000 lensing rotation curves of isolated galaxies [1]. They span 0.035–2.6 Mpc in four stellar-mass bins ($\log M_\star = 8.5$–11), with the full 60×60 covariance and the multiplicative bias applied.
- **Fitting.** Baryonic masses are profiled per bin, with the same freedom for every model.
- **Ranges.** The primary range is $R \leq 0.31$ Mpc, where the isolated sample is clean. All radii are secondary.
- **Footings.** $a_0 = 9.36\times10^{-11}$ (canonical) and $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$ (alternative), never pooled.

2 Results

model	$\Delta\chi^2$ vs edgeless law ($R \leq 0.31$ Mpc)	all radii
edge at $5.84\,r_M$ (own share only)	$+197$	$+753$
infall-filled law (no edge)	$+0.1$	$+2.8$
census catchment edge ($N = 1/f_{\rm ret}$, no fit)	-0.2	$+149$
census catchment + framework two-halo (vs edgeless + two-halo)	-2.1	$+11$

Controls. Every lane passed its reproduction checks. Each deliberately broken variant reproduced the failure it should, except two:

- In the two-halo lane, a σ_8 cross-check failed by 3.2% (a 2% tolerance). The amplitude is absorbed by the profiled bias.
- The no-exclusion variant shifted the bias in only two of the four bins.

3 Interpretation and limits

- **Where the cold mass is.** The data rule out a cold halo confined to each galaxy's own cosmic share. They are consistent with cold energy gathered from a larger catchment by collisionless infall, which is the framework's own dynamics.
- **The supply factor is an input.** N comes from an observational baryon census. A framework-native derivation was attempted and failed: an energy-budget model retains almost all baryons.
- **Inherited simulation inputs.** The two-halo term uses a simulation with inherited ΛCDM initial conditions. A native-input rerun is queued.
- **The absolute fit is poor for all models.** The heavy bins dominate it: the inner points are over-predicted and the 0.2–1.4 Mpc points under-predicted. This is the massive and early-type excess, which the framework does not yet explain.

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[1] M. M. Brouwer et al., A&A 650, A113 (2021) (KiDS-1000 weak-lensing RAR; public ESD profiles and covariance). [2] This programme (public repository): lanes CFG801 (edge), CFG803 (infall), CFG804 (census catchment), CFG805 (two-halo), CFG806 (census derivation, not derived), and ARGON02/AG4 (settled profile from the action). Each lane has frozen criteria committed before its script. [3] T. Mistele, S. McGaugh, F. Lelli, J. Schombert, P. Li, ApJL 969, L3 (2024). AI assistance: Anthropic's Claude; parallel AI sessions.